Duality

$\min f(x)$ s.t. $h(x)=0$, $\varphi(x)\le 0$ (P)

Lagrangian func $L = f + \lambda h + \mu\varphi$

Dual func $g = \min_x L$ s.t. $\mu \ge 0$

g must concave func Pf：线性 + 逐点上界

Dual problem

$\max g(\lambda, \mu)$ s.t. $\lambda \ge 0$, $\mu \ge 0$ (D)

(D) must convex problem

$g(\lambda) \le d^* \le p^* \le f(x)$

Pf: $d^* = \max_\lambda g(\lambda) = \max_\lambda \min_x L(x,\lambda) \le \max_\lambda L(x^*,\lambda)$

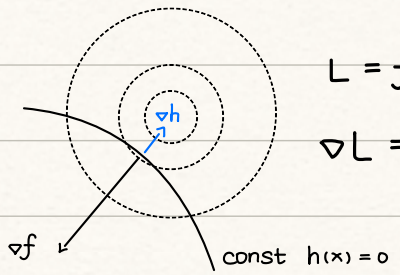
↳ 每个x关于λ，不是x*

$p^* = f(x^*) = L(x^*, \lambda)$

与λ无关，反正 $h=0$

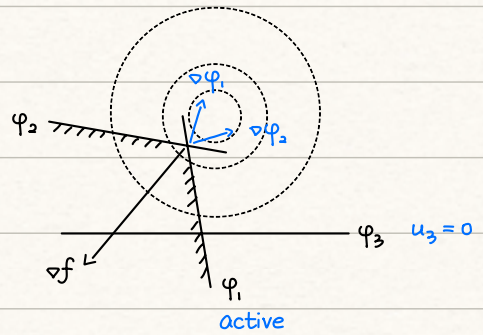
KKT 1st NOC

$\nabla L = 0$, $h(x)=0$, $\mu\varphi(x)=0$, $\mu \ge 0$, Regular



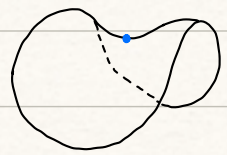
$L = f + \lambda h$

$\nabla L = \nabla f + \underbrace{\lambda}_{\text{倍数}} \nabla h = 0$

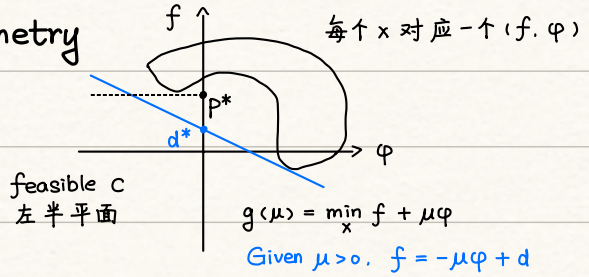


Saddle point $L(x,\lambda^*) \ge L(x^*,\lambda^*) \ge L(x^*,\lambda)$

Strong Dual $\max_\lambda \min_x L = \min_x \max_\lambda L$ $d^* = p^*$ $\Updownarrow$

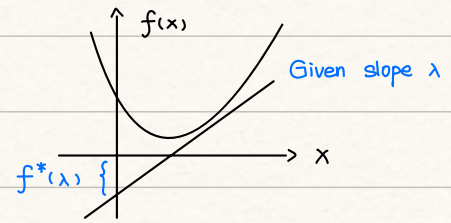


Geometry



Conjugate

$f^*(\lambda) = \max_x \ \lambda^T x - f(x)$



e.g. $f(x) = ax + b$, $f^*(\lambda) = -b$ 定义域仅 $\{a\}$

$I_C^*(\lambda) = \max_{x \in C} \langle \lambda, x \rangle$ Support func

e.g. $C = [0,1]$, $I_C^*(\lambda) =$

Prop. f^* 必凸 Pf：逐点上界必凸

$f^{**} = f$ 若凸闭

Fenchel $f(x) + f^*(\lambda) \geq \lambda^T x$

conjugate 和 dual

min $f(x)$ s.t. $x = 0$ 则 $-g(\lambda) = f^*(-\lambda)$ 奇对称

Pf：$L = f + \lambda^T x$, $g(\lambda) = \min_x L = \max -\lambda^T x - f(x) = -f^*(-\lambda)$